

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-CA-010

Title: Management of PLC/DCS Logic Changes and Interlock Bypass

Department: Control and Automation

Revision: 01

Date: 16-Jul-2024

1. PURPOSE

To ensure that changes to PLC/DCS programs and any bypass (forcing) of safety interlocks are authorised, recorded and removed in time.

2. SCOPE

Applies to all PLC, DCS and SCADA systems maintained by the Control and Automation (C&A) Department, including level-2 systems.

3. LOGIC CHANGE PROCEDURE

Step 1 — Raise a Logic Change Request (form CA-LCR-01) describing the change and the reason.

Step 2 — Get approval from the Area Head of C&A and the Operations Head of the concerned shop.

Step 3 — Take a full backup of the existing program and store it on the C&A backup server with date and version.

Step 4 — Test the change in simulation mode where available.

Step 5 — Download during a planned stoppage, verify, and update the change register.

4. INTERLOCK BYPASS (FORCING)

- Safety interlocks (e.g. gas pressure low trip, lance water flow low trip, crane over-travel) may be bypassed only with a written Interlock Bypass Permit
- Maximum bypass duration: 8 hours (one shift); extension needs fresh approval
- A red "INTERLOCK BYPASSED" tag must be placed on the HMI screen and on the field device
- An alternative safety measure (e.g. manual watch) must be in place for the whole duration
- All active bypasses must be reviewed at every shift handover

5. PASSWORDS AND ACCESS

- Engineering station passwords are changed every 90 days
- USB drives may be used only after virus scanning on the C&A clean station

6. RECORDS

All changes and bypasses are recorded in the C&A Change Register and kept for 3 years.